CHARACTERIZATION OF SUPRAHARMONIC EMISSION FROM HIGH-POWER ELECTRIC VEHICLE CHARGERS

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ABSTRACT

The increasing number of high-power electric vehicle (EV) charging stations in electricity grids introduces new challenges for the grid operators. Not only the high power demand of the chargers has an impact on the grids, the devices can also be a source of supraharmonic (2-150 kHz) emissions that have several unwanted effects. In this research, an advanced set-up is developed to characterize the supraharmonic emission of high-power EV chargers to find the influencing parameters. Two chargers with a 175 kW rating are tested on a wide range of DC-voltage and -current setpoints and for different AC grid conditions to obtain a representative assessment of the behaviour of the chargers under different operation conditions. The main conclusion is that the supraharmonic emission of the chargers is highly dependent on the DC output voltage and that multiple chargers may interact with each other, resulting in a slow time-varying beating phenomenon. The findings support the understanding of the supraharmonic emission from EV fast chargers and help in assessing the impact they can have on the grid.

1. INTRODUCTION

The number of high-power or fast-charging stations (FCSs) for electric- vehicles, busses and trucks (EVs) in distribution grids is developing to accommodate the growing demand for a comprehensive fast-charging infrastructure [1]. Due to their high power demand, typical locations have multiple units of up to 350 kW to avoid long charging times, FCSs can have an impact on the grid and require a strong grid connection [2]. Additionally, FCSs are large switching power electronic devices and influence the quality of power supply as they can be a source of harmonic emission and cause resonances [3]. Furthermore, FCSs can be a source of supraharmonic (SH) emissions [4], [5]. Supraharmonics are waveform distortions in voltage and current between 2 and 150 kHz, and are observed in low- and medium voltage grids [6]. They can result in several unwanted effects and interference with customer equipment and network components [7].
The effect of SHs from FCSs is still under investigation, and it is unknown how one or multiple FCSs can contribute to SH emissions in the grid and which factors influence the emission frequency and amplitude. This information is of paramount importance for a grid operator, who should be able to investigate potential effects prior commissioning, to avoid or mitigate any problems in the installation. This facilitates the smooth integration of emission free transport in our electricity system.
This research aims to provide a characterization of the supraharmonic emission from FCSs by testing the chargers for different AC grid conditions and performing a sensitivity analysis of the charger parameters. For the chargers, the DC voltage and current setpoints are varied and different output power settings are obtained.
The contributions of this research are as follows:

- Presentation of the developed test set-up.
- Sensitivity analysis of the SH emission to the DC-voltage, DC-current, Output power, AC-grid conditions.
- Assessment of SH interaction

In Section 2, the test set-up and the chargers under study will be presented. Section 3 will discuss the results regarding the SH emission from the chargers and the factors influencing it are determined. Interaction between the chargers is analysed in Section 4 and the most important conclusions are given in Section 5.

2. METHODOLOGY

In this research, 2 large FCSs rated 175 kW are experimentally tested in a controlled lab environment. Using an advanced setup with control over both the chargers and the grid conditions, the FCSs are tested for various scenarios. Using this method, a quantitative assessment of the FCS impact is provided, and using sensitivity analysis the different operation states of the device are assessed separately. The FCSs operate on a low-voltage 400V line-to-line network that is either supplied by a dedicated MV/LV transformer or via a 500 kVA grid emulator, see Figure 1. The parameters of the chargers that were controlled are: DC-voltage (between 460-800Vdc), output power (between 50-175 kW), and output current. During all tests, the voltage and current waveforms are measured using an oscilloscope / waveform-recorder with a 16-bit resolution, a sample rate of 1 MS/s, and bandwidth of 300 kHz. The voltage is measured using 10:1 passive probes, and the current using flexible Rogowski coils with a bandwidth of 600 kHz (-3 dB) and 10 mV/A resolution. The data is post-processed and analysed using MATLAB. The calculation and grouping method for the supraharmonic components will be further elaborated in this section.

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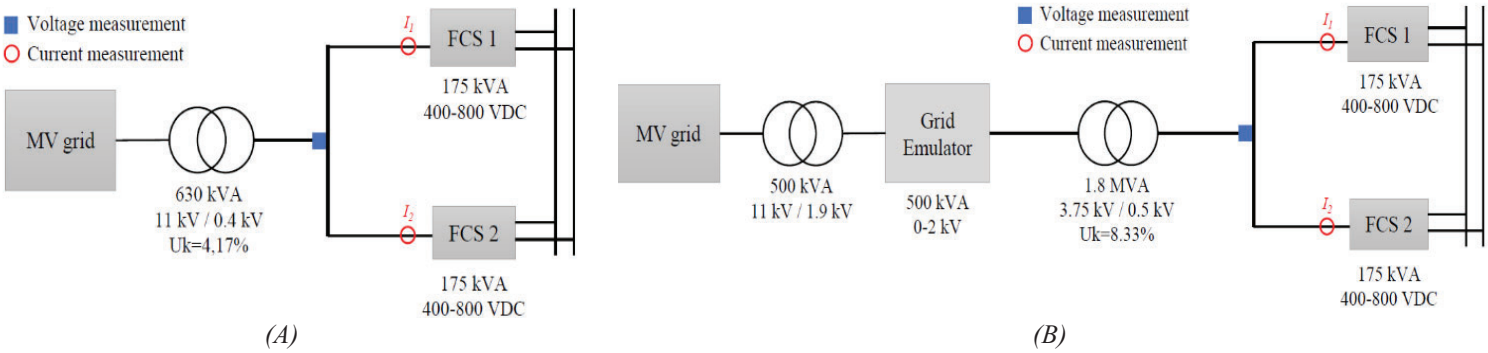


Figure 1. Measurement set-up, with chargers connected back-to-back directly through a dedicated MV/LV transformer *(A)* or via a grid emulator and isolation transformer *(B)*. Voltage and current measurements points are indicated.

Test set-up

This section will provide an overview of the different components of the set-up. The measurement set-up is in the Flex Power Grid Lab (FPGL) of KEMA laboratories in Arnhem, The Netherlands. Two different set-ups are used, as presented in Figure 1. The chargers are denoted as FCS 1 and FCS 2 and are connected back-to-back to the same LV network. In this set-up, FCS 1 will act as a source (AC to DC mode) whereas FCS 2 will act as sink (DC to AC mode). The measurements are repeated with these settings reversed; FCS 2 as source and FCS 1 as sink. This due to unavailability of 2 electric vehicles (EV) with sufficient high charger power and battery capacity. However, when using an EV, the state of charge, battery capacity and other mainly uncontrollable parameters will determine the voltage and current setpoint, which makes obtaining reproducible test results difficult. In this set-up, there is more controllability compared to using an actual EV as the voltage and current setpoints can be controlled freely and can be contained for an indefinite amount of time. The measurements will be based on the combination of the two chargers operating simultaneously.

In set-up A (Figure 1-A), the low-voltage network is supplied by a distribution 11 kV / 400 V transformer of 630 kVA rating. This offers a more realistic connection for the chargers, mainly because the impedance of the transformer is significantly lower. However, the voltage parameters cannot be controlled and distortion from neighbouring labs might be present on the upstream MV that propagates to the LV.

The low-voltage network is in set-up B (Figure 1-B) supplied by a 500 kVA grid emulator that offers full control over the voltage waveform and eliminates background distortion from the upstream MV grid by building a clean voltage. Harmonics can be added up to 2 kHz and voltage parameters per phase are controlled. An 1.8 MVA isolation transformer is used to convert the output of the grid emulator to 400 V line to line and serves as isolation to avoid any interaction between the control loops of the separate systems.

High-power chargers

Two FCSs of similar type, are used for the tests. The specifications of the chargers are presented in Table 1.

Table 1 Charger specifications

Max DC output power	175 kW
Max DC output current	350 A
Output DC voltage range	460-800 V
Power factor	> 99 %
Power conversion efficiency	> 96 %

The chargers have a rated power of 150 kW over the complete DC voltage range. For a DC voltage around 600V a higher power of 175 kW is possible. DC voltage output ranges from 460 to 800 V. The DC output voltage and DC output current values are controlled via a development interface.

DC voltage and current setpoints

Different combinations of output power level and DC output voltage are obtained according to the 9 different setpoints as shown in Table 2. The chargers operate on each setpoint for 15 minutes and a measurement snapshot is taken every 10 seconds. Only for 600Vdc the power of 175 kW is obtained, for 460 Vdc and 800 Vdc the maximum power is 150 kW due to charger limitations. Three steps in power are used, representing 50, 100 and 150 or 175 kW output power. A power flow is initiated by decreasing the voltage of the charger that acts as a sink slightly. Communicated protocols (CCS) are overruled and the different steps (e.g., pre-charge, closing contactors) to initiate the charging are performed manually using the interface.

Table 2 DC voltage and current setpoints

	50 kW	100 kW	150/175 kW
460 Vdc	108 Adc	217 Adc	336 Adc
600 Vdc	83 Adc	166 Adc	292 Adc
800 Vdc	63 Adc	125 Adc	188 Adc

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Supraharmonic calculation

The 5 Hz frequency components I_f are obtained with the Fast Fourier transform function of MATLAB, with a 200 ms window. The resulting components are then grouped in supraharmonic bands I_{sh} of 1 kHz width as

$$I_{sh} = \sqrt{\sum_{f=f_{min}}^{f_{max}} I_f^2}$$

obtaining 148 bands. Similarly, the voltage components are grouped in V_{sh} bands. The SH emission frequencies of the chargers under study are identified in the 3-4 kHz and 7-8 kHz bands and therefore the focus will be on these bands only, as no additional emission frequencies were identified during the lab tests. The emission spectra of the chargers are as presented in past research [5]. In the next 2 sections the results based on the tests with set-up A, grid connected, are presented.

3. RESULTS AND CHARACTERIZATION

Emission in 3-4 kHz band

For the emission in the 3-4 kHz band, the results are described in this subsection. The emission with respect to the DC voltage is presented in Figure 2. For each subplot the power level is fixed (50, 100, 150 kW) and the DC voltage is varied. It is observed that for an increasing DC voltage, there is a slight decrease in emission in this band. The spread is however quite wide, the emission ranges from approximately 5 to 13 A (75th percentiles). When analysing the mean values, a decrease in emission is observed for each increase in DC-voltage. The relation, however, is not constant and differs depending on the output power.

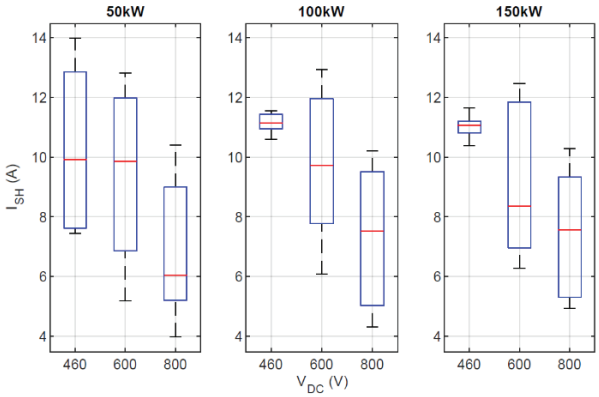


Figure 2 SH emission in 3-4 kHz band vs. DC voltage, for different output power.

In Figure 3 the mission in the 3-4 kHz band is plotted with respect to output power instead of DC voltage. The DC voltage is fixed for the different subplots, and the output power is varied. Again, the range in which the emission amplitudes occur is quite wide (from 5 to 13 A based on the 75th percentiles) and there is no clear relation between the emission and the output power.

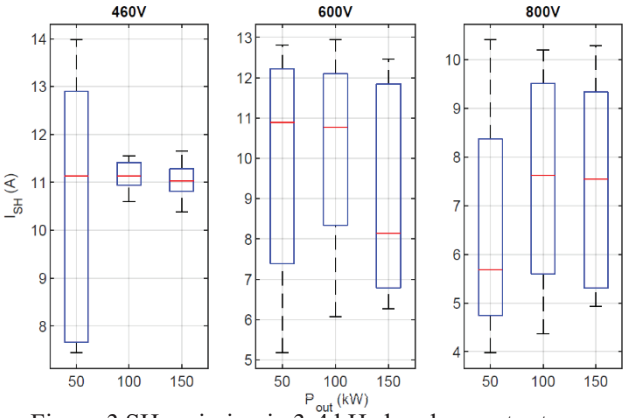


Figure 3 SH emission in 3-4 kHz band vs. output power, for different DC voltages

Based on the mean values, the emission increases first for 460 V and then decreases for increasing power. For 600 V the emission decreases for increasing power, and for 800V it increases and then decreases. No clear relation is found between the emission in the 3-4 kHz band and the output power, and some relation with the output DC voltage, but due the large overlap in values not confident enough to draw conclusions.

Emission in 7-8 kHz band

Dominant emission is also observed in the 7-8 kHz band. The SH emission with respect to the DC voltage is presented in Figure 4. Note that the y-axes are linked in this figure. It is observed that the emission is similar for different output power, and highly dependent on the DC output voltage of the chargers. For a DC voltage of 460 V the emission is the lowest (between 0.5 and 0.7 A). With increasing DC voltage, the emission in this band also increases. For 600 V the emission approximately doubles and is between 1.2 and 1.4 A. For a further increased voltage of 800 V the values double again compared to the 600 V values. The emission is now between 2.2 and 2.4 A.

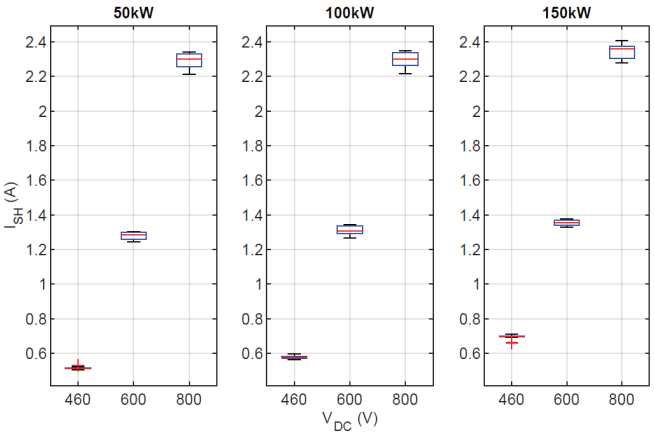


Figure 4 SH emission in 7-8 kHz band vs. DC voltage, for different output power.

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For the different power levels, the emission is similar. When analysing the emission with respect to the power level, Figure 5 is obtained. Note that the y-axes are not linked in this figure, for improved readability of the small differences. It is observed that the emission in the band is also slightly dependent on the output power of the chargers. The effect is less compared to the output voltage, an increase of maximum 0.2 A is observed when fixing the DC voltage and increasing the power level. This implies that the relative emission, as scaled to the fundamental current, will decrease for an increasing power, whilst the absolute values will slightly increase.

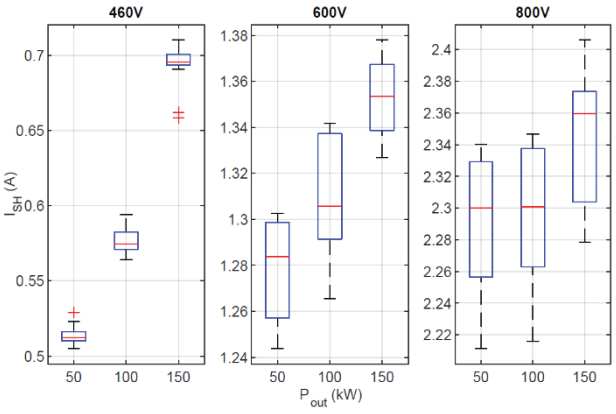


Figure 5 SH emission in 7-8 kHz band vs. output power, for different DC voltages

Effect of grid emulator (set-up B)

All tests are repeated in set-up B, where the low-voltage network is supplied by the grid emulator. The same voltage and current setpoints as presented in Table 2 are used for the tests. This section will discuss the similarities and differences between the tests with set-up A and set-up B.

Regarding the emission of the chargers, the same SH emission frequencies are identified. For the 3-4 kHz band, the emission shows the same behaviour as presented in Figure 2 and 3 with almost identical emission amplitudes for the different setpoints. Hence, this result is not shown in a figure. The same holds for the emission in the 7-8 kHz band, an identical behaviour is observed for the different voltage and current setpoints, with similar emission amplitudes and the similar relation between emission and input power or DC-voltage is observed as in Figures 4 and 5. Hence this result is not shown here.

The main difference between the emission in set-up A and set-up B is the effect on the voltage. Both set-ups will give a different apparent upstream impedance, that is defined by the MV/LV transformers' short circuit rating, and the upstream elements (MV grid or grid emulator). Due to this, a similar current emission results in a different effect on the voltage.

For the 7-8 kHz band and 150 kW power this difference is illustrated in Figure 6. On the left part of the figure the SH voltage disturbance in the 7-8 kHz band is presented for set-up A (grid connected) and in the right part of the figure for set-up B with the grid emulator (GE) included.

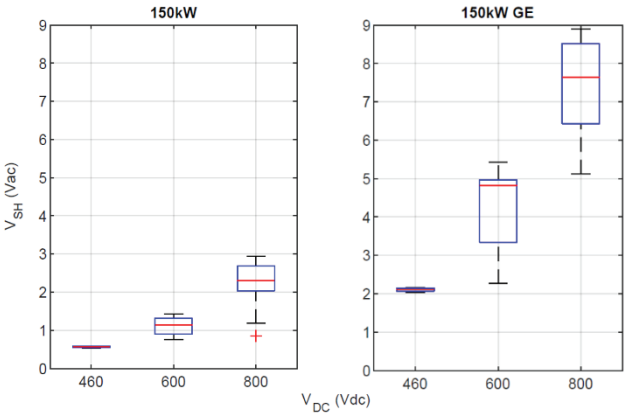


Figure 6 SH Voltage in 7-8 kHz band vs. DC-voltage, for grid connected (left) and grid emulator (right) set-ups and an output power of 150 kW

It is observed that the disturbance on the voltage shows a similar trend as in Figure 4, for an increasing DC-voltage, the disturbance increases. For the steps 460 to 600 V and 600 to 800 V the voltage disturbance almost doubles, and ranges between 0.5 and 3 V for set-up A.
For set-up B, the SH voltage disturbance shows a similar trend, but at a higher amplitude. Note the y-axes that are linked for easy comparison. Whereas on the grid connection the voltage disturbances remain below 3 V, for the grid emulator set-up the disturbances increase up to 9V. In general, the SH voltage in the 7-8 kHz band is 3 times higher in test set-up B.

This is an important finding that shows the effect of the grid impedance on the voltage disturbances. A higher grid impedance will result in a higher voltage disturbance for a similar emission. Whilst the fundamental or short-circuit impedance of the transformers is similar, the impedance for higher frequencies will highly depend on the X/R ratio of the transformers. The short circuit or equivalent impedance Z_{sc} of a transformer is either given or can be derived from an otherwise specified short-circuit voltage ratio U_k. This ratio is expressed as the percentage of the nominal voltage U_N needed on the primary side to achieve the nominal current of the transformer on the secondary side, given that the secondary side is in a short-circuit condition. The frequency dependent impedance of a transformer can be estimated by scaling the inductive part of the transformer short circuit impedance with frequency as

$$Z_{est}(f) = |R_{sc} + j \cdot f \cdot \frac{X_{sc}}{50}|$$

with frequency f and R_{sc} and X_{sc} the short-circuit resistance and the short circuit reactance respectively [8].

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4. INTERACTION

As observed in the results for the 3-4 kHz band, the emission amplitude stretched across a wide range. This is further analysed by taking snapshots of a longer duration, at a lower sample rate and a shorter FFT window (affecting the amplitudes). It is found that the emission in this band is time-varying. The result in shown in Figure 7 for the setpoint 100 kW at 600 Vdc. The emission varies with a period T of approximately 0.5 seconds, which might be due to the emission from FCS 1 and FCS 2 that are at approximately the same frequency, resulting in an additional component at the difference of the frequencies according to $f_{beat} = |f_1 - f_2|$ as described in [9], in this case around 2 Hz. This slight difference in switching frequency of the individual converters results in an amplitude modulation of the emission in the 3-4 kHz band. This interaction should be considered when assessing the emission from installations with similar devices emitting supraharmonics.

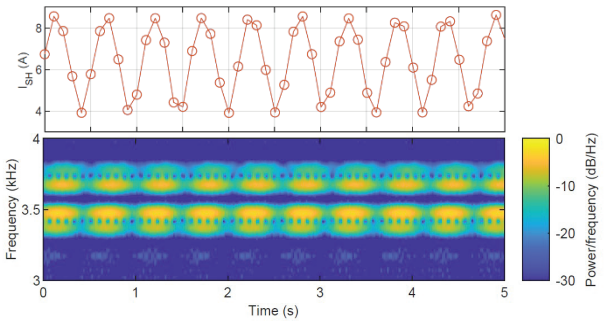


Figure 7 SH emission in 3-4 kHz band, amplitude (top) and spectrogram (bottom), 100 kW/600 V setpoint, 5 seconds.

5. CONCLUSION

In this research the factors that influence the supraharmonic emission from fast charging stations are studied. A test set-up is developed and different voltage and current setpoints are used to obtain a representative insight into the emission of the chargers for the operating range. It is found that part of the SH emission is highly sensitive to the DC-voltage, which shows a linear relation. The output current or, resulting output power has a lesser impact on the emission, but further research on a different test set-up should conclude on this matter.

The supraharmonic emission from the two or more identical chargers (or other devices) can cause interaction, resulting in a slow, time-varying emission characteristic.

The grid connection is found to influence the supraharmonic voltage distortion due to the different frequency dependent network impedance. The supraharmonic current components remain mostly constant in amplitude. Further research should focus on the characterization of the high-frequency impedances of transformers and other network components, to make a quantitative assessment of the expected voltage disturbance that can be caused by an installation emitting supraharmonics.

Due to the interaction, no quantitative characterization could be made for the emission in the 3-4 kHz band. Future test should take this into account and preferably use separated grid connections to avoid this issue. However, in a real grid situation with multiple chargers connected, it is expected that the same effect will occur. This should be considered when assessing the maximum emission from an installation with fast charging stations.

ACKNOWLEDGMENTS

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